

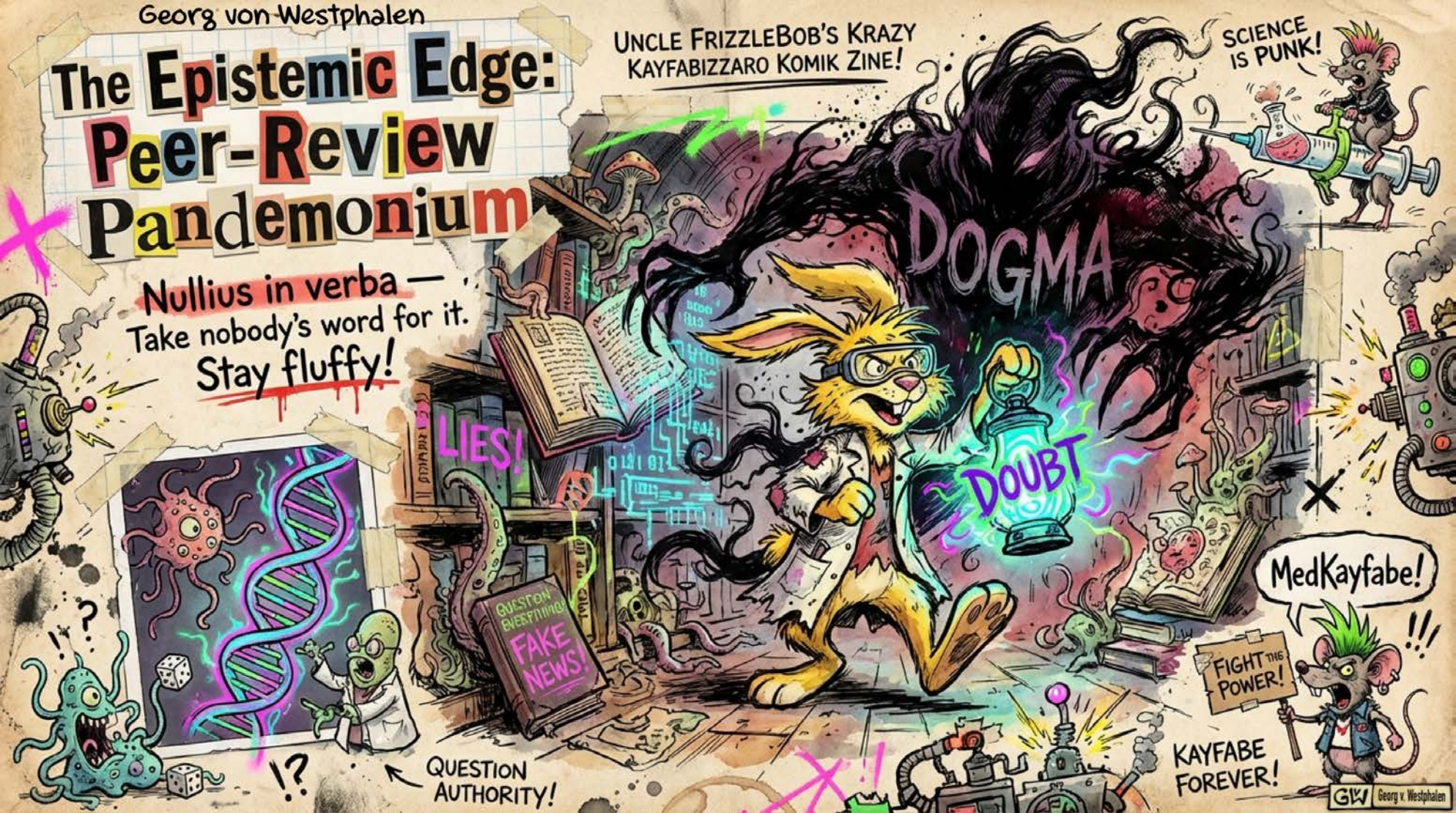
Georg von Westphalen

The Epistemic Edge: Peer-Review Pandemonium

Nullius in verba —
Take nobody's word for it.
Stay fluffy!

UNCLE FRIZZLEBOB'S KRAZY
KAYFABIZZARO KOMIK ZINE!

SCIENCE
IS PUNK!



LIES!

DOUBT

QUESTION EVERYTHING!
FAKE NEWS!

MedKayfabe!

FIGHT THE
POWER!

KAYFABE
FOREVER!

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Alhazen-the-Observer

"The Method"

Year 1021

Book of Optica

First Systematic Method



POWER (The Skeptic's Lens): Demands every Theory-Wrestler prove themselves through repeatable experiments, not elegant words.
LORE: The father of the Scientific Method. He moved science from Philosophical-Guessing to Empirical-Testing.

Semmel-the-Outcast

"The Ignored Signal"

1847

Puerperal Fever

90% reduction in Death-Stats Ignored



POWER (The Tragic-Signal): Identifies the Invisible-Enemy before the arena is ready to believe it, suffering professional exile.
LORE: The cost of being right too early. Evidence is often rejected by the Consensus-Guard if it challenges their status.

KAYFABIZZARO!

Vesali-the-Dissector

"Anatomy of Truth"

1543

De Humani Corporis Fabrica
Overturned Galen



POWER (The Dogma-Smasher): Physically shows that the Old-Scrolls are wrong by looking at the Real-Wrestler-Body.
LORE: Observation trumps tradition. If the book says one thing and the body shows another, trust the body.

Diagnosis?!



Pasteur-the-Vaccinator

"Germ Theory"

Germ Theory

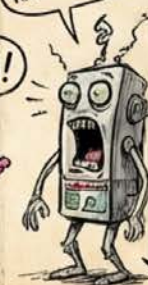
Pasteurization

Rabies success 1885



POWER (The Micro-Defense): Proves that Invisible-Invaders cause the Arena-Collapse, providing the first systematic Shields.
LORE: The bridge from basic biology to clinical intervention. Turned Miasma-Mystery into Pathogen-Logic.

TRUTH!



ANARCHY!

UNCLE FRIZZLEBOB!

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Popper-the-Falsifier

Karl Popper

STAT:
The Logic of Scientific Discovery
Falsifiability Criterion

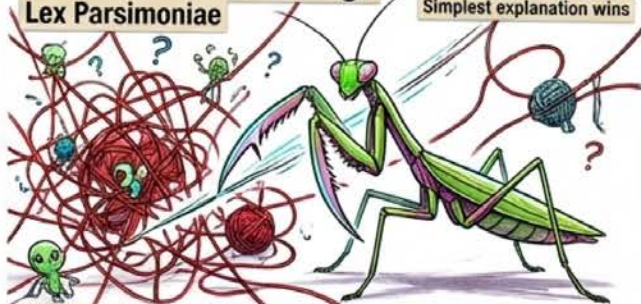


POWER (The Black-Swan-Strike): Declares a Theory-Wrestler is only scientific if there is a move that can Kill It.
LORE: Science doesn't Prove truth; it Eliminates falsehood. A theory stays in the ring only until falsified.

Ockham-the-Razor-Edge

Lex Parsimoniae

STAT:
Simplest explanation wins

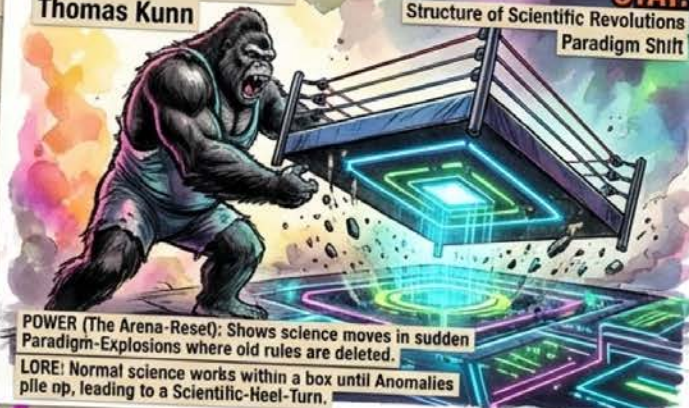


POWER (The Efficiency-Cut): Slices through complex, bloated Conspiracy-Theories to reveal the simplest path fitting the Stat-Data.
LORE: Entities should not be multiplied without necessity.

Kuhn-the-Shifter

Thomas Kunn

STAT:
Structure of Scientific Revolutions
Paradigm Shift



POWER (The Arena-Reset): Shows science moves in sudden Paradigm-Explosions where old rules are deleted.
LORE: Normal science works within a box until Anomalies pile up, leading to a Scientific-Heel-Turn.

Scientism-the-False-God

The Critique

STA:
Dogmatic adherence to Consensus
Rejection of doubt

POWER (The Dogma-Daze): Wears the mask of Science and Behaves like a Religion, silencing Doubt-the-Flat.
LORE: The corruption of the method. Scientism claims science has All the Answers, forgetting It is the Process of Guessing.



KayfaBIZARRO!

RCT-the-Gold-Standard

Randomized Controlled Trial

STAT: Phase I-III
Double-Blind
Placebo-Control



POWER (The Bias-Shield): Ensures the Chemical-Ally wins because it's socially strong, not because the coach believed it.
LORE: Randomization and Blinding. The only way to defeat Placebo-the-Trickster.

P-Value-the-Probabilist

Statistical Significance

STAT: $p < 0.05$
Significance threshold



POWER (The Fluke-Filter): Tells you if the Match-Outcome was a Real-Skill-Win or a Random-Dice-Roll.
LORE: Probability of observing results if the Null is true. Caveat: Easily bribed by P-Hacking-the-Cheat.

Null-the-Boring-Ghost

Null Hypothesis

STAT: H_0
The assumption of No Effect



POWER (The Status-Duo-Guard): Assumes the new Drug-Mixer is useless until proven otherwise with Overwhelming-Evidence.
LORE: The conservative heart of sciences. You don't win until you Kill-the-Ghost.

Meta-the-Master-Review

Meta-Analysis

STAT: Forest Plot
Systematic Review
Level 1 Evidence



POWER (The Arena-Synthesis): Gathers 50 different Match-Tapes and combines them into one Giant-Verdict.
LORE: Pooling data to increase power. Identifies the True-Signal buried in the Noise of smaller matches.

Arena Ring

Call Doc
Frizzle!

SCIENCE PUNK!

RESIST!

Big-Pharma-the-Lobbyist Corporate Science

STAT: \$2B to \$3B
average cost per drug
Marketing > R&D



POWER (The Funding-Friction): Decides which Wrestlers get into the arena based on Profit-Potential rather than Cure-Potential.
LORE: Research follows the money. Creates Incentive-Glitches treating only profitable symptoms.

P-Hacker-the-Cheat Data Torture

STAT: Cherry-picking
Post-hoc analysis
Over-testing



POWER (The Statistic-Torture): Tortures the Data-Wrestler until R Confesses to a significant result, even if there isn't one.
LORE: Running 100 tests until one looks Significant by chance. Turns Noise into False-Verdicts.

Publicat-the-Bias-Guard Publication Bias

STAT: Positive results are 3x
more likely to be published



POWER (The File-Drawer Trap): Hides every Losing-Match-Tape in s basement, creating the illusion of a 100% win rate.
LORE: The Filter-of-Success. It skews Meta-Analysis because the Evidence-Pile is missing all failures.

Conflicts-the-Shadow COI

STAT: Undisclosed funding
Industry-sponsored trials



POWER (The Hidden Influence): Whispers to the Principal-Investigator, interpreting a Tie-Match as a Victory-for-the-Sponsor.
LORE: Financial ties compromise objectivity.

Arena Ring

Call Doc
Frizzle!

HUMBURG!

SCIENCE PUNK!

RESIST THE
SYSTEM!

AI-the-Pattern-Predictor Modern Research

STAT: Neural Networks
AlphaFold / Big Data



POWER (The In-Silico-Sprint): Runs a billion Virtual-Matches in a second to find the perfect Protein-Armor-Shape before a real lab opens. **LORE:** Machine learning in drug discovery. Finds patterns the human brain could never see.

Precision-the-Genetic-Sniper Personalized Medicine

STAT: Pharmacogenomics /
N=1 Trials



POWER (The Individual-Verdict): Tailors the Chemical-Ally to fit your specific DNA-Lock. **LORE:** Moving from Population-Average to Individual-Reality. The Standard-Match is replaced by the Personal-Match.

Pre-Print-the-Speedrunner Rapid Dissemination

STAT: ArXiv / BioRxiv
Zero Peer-Review



POWER (The Information-Burst): Bypasses the Peer-Review-Pandemonium to deliver Match.Results instantly. **LORE:** Rapid dissemination vs Rigor. Great for Pandemic-Panic, but prone to Error-Glitches.

Open-the-Access-Knight Democratic Data

STAT: CA Journals / Sci-Hub
Public-funded research



POWER (The Paywall-Breaker): Smashes the Journal-Fortress gates so every Rookie-Doctor can see the Research-Scrolls. **LORE:** The movement for Democratic-Data. If the public paid for the match, they should see the tape for free.

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OBSERVATIONAL

Case-Control-the-Flashback Retrospective

STAT
Odds Ratio (OR)
Backward in time



POWER (The History Search): Finds wrestlers who already lost, looking back to see if they touched the same Yoshi-Vandal.
LORE: Outcome back to Exposure. Best for Rare-Zebra hunts where you can't wait for a new match.

Cohort-the-Follower Prospective

STAT
Relative Risk (RR)
Forward in time



POWER (The Long-Watch): Takes a group of Exposure-Wrestlers and follows them for 20 years to see who Crashes first.
LORE: Exposure forward to Outcome. The Gold Standard of observational matches.

BLÖDSINN!



INTERVENTIONAL

Double-Blind-the-Justice Trial Control

STAT
Neither doctor nor
patient knows treatment



POWER (The Bise-Erser): Ensures Expectation-Sparks don't pollute the Data-Stream. Both Wrestler and Ref are in the dark.
LORE: The ultimate defense against Placebo-the-Trickster and Investigator-Effort-Bias.

KAYFABIZZARO!

Placebo-the-Trickster The Baseline Control

STAT
Sugar pill
The Expectation-Effect



POWER (The Mind-Healer): Heals the Wrestler just by making them believe they got a Shemp-Chemical-Ally.
LORE: The power of belief. Science must prove its Move is stronger than the Trickster's noise.

SUNK-ROCK!

BLÖDSINN!



Belmont-the-Protector Research Ethics

STAT:
1979 Report
Beneficence,
Justice, Respect



POWER (The tttman-Shield): Ensures Wrestlers are not treated like Laboratory-Rats and the match is worth the risk.
LORE: The response to Historical-Horrors. Research must serve the subject, not just the Data-Gods.



Stay
fluffy!



Informed-the-Consent-Scroll Autonomy

STAT:
Right to withdraw
Full disclosure



POWER (The Sovereign-Voice): Ensures the Wrestler knows exactly what Monsters are in the arena before entering the ring.
LORE: The legal and ethical Shield of Autonomy. Science cannot happen in secret.



IRB-the-Gatekeeper Ethics Board

STAT
Institutional Review Board
The Pause button



POWER (The Protocol-Police): Reads the Match-Plan before it starts and Blocks any match that is Logicality-Shaky.
LORE: Peer oversight. Prevents Mad-Scientist-Heel-Turns by demanding ethical accountability.

$$N_{in} = \binom{n}{x=0} a + (n-1)^2$$

What
theFLUFF?

Declara-the-Helsinki Global Ethics

STAT
WMA Declaration
Revised internationally



POWER (The Universal-Rulebook): Sets the Global-Standard for how Human-Wrestlers must be treated in every arena on Earth.
LORE: The ethical Constitution of the World Medical Association.



SUNK-ROCK!



What the FLUFF?!

$$N_{in} = \binom{n}{x=0} a + (n-1)^2$$



OBSERVATIONAL & INTERVENTIONAL (The Data-Kayfabe)

Big-Data-the-Giant
Real-World Evidence

STAT
Terabytes of data
Total Observation



POWER (The Infinite-Library): Looks at millions of Electronic-Health-Scrolls to find Hidden-Patterns a tiny RCT would miss.
LORE: The sluff from Sampling to Total-Observation. Powerful, but prone to Noise-Glitches.

Registra-the-Trial-Tracker
Trial Registration

STAT
ClinicalTrials.gov
Preventing File-Drawer bias



POWER (The Public-Vow): Makes the Coach announce the Match-Rules before the fight, preventing them from changing the Winning-Criteria later.
LORE: Transparency. Prevents Outcome-Switching and Harking.

Confound-the-Distractor
Confounding Variable

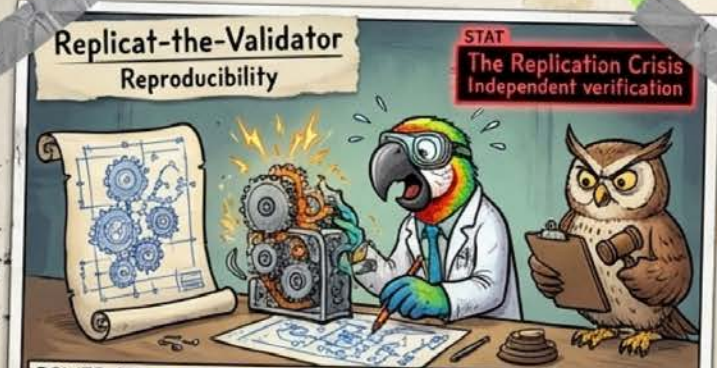
STAT
Third variable distorting
the relationship



POWER (The Illusion-Mist): Makes it look like Move-A caused Outcome-B, but really it was Variable-C hiding in the shadows.
LORE: Correlation is not Causation. The nemesis of Logic-the-Razor.

Replicat-the-Validator
Reproducibility

STAT
The Replication Crisis
Independent verification



POWER (The Reality-Check): Demands the Miracle-Victory be performed again in a Different-Arena. If it can't be repeated, it was a lie.
LORE: A single win is a fluke; a repeatable win is a fact.



Petri-the-Culture-Dish In Vitro

STAT: Cell lines
Controlled environment



POWER (The Isolated-Brawl): Allows the Chemical-Ally to fight the Pathogen-Punk in glass without the Wrestler-Body interfering.
LORE: The Initial-Testing-Ground. Caveat: Cells in a dish don't behave like cells in a human.

Murine-the-Mouse-Model Animal Research

STAT: Translational research
Genetic knockouts



POWER (The Proxy-Wrestler): Takes the Chemical-Hits first to see if the Arena-Stats are safe for humans.
LORE: Translational bridge from Dish to Human, though Mice are not small men.

Western-the-Protein-Blot Protein Quantification

STAT: SDS-PAGE
Target verification



POWER (The Protein-Scanner): Identifies exactly which Protein-Armor-Pieces are being built by the cell and in what quantity.
LORE: Molecular biology. The Hardware-Audit of the cell.

Knock-Out-the-Deletion Gene Silencing

STAT: CRISPR
Targeted gene disruption



POWER (The Source-Code-Kill): Deletes a single Manual-Page (Gene) to see if the Wrestler can survive without it.
LORE: Functional genomics. If the wrestler Crashes, we know the gene's Power-Level.

GRANT APPLICATION
REJECTED!

SCIENCE PUNK!

SUNK-ROCK!

STAY FLUFFY!

Grant-the-Fuel-Source Research Funding

10-20% success rate
Public funding (NIH/ERC)

POWER (The Lifeline):
Provides the Tokens needed to pay for the Arena and the Wrestlers. Without him, science stops.

LORE: Competition for resources. Only the most Buzzword-Heavy and Safe matches get funded.

Nah heh! Peer Review tax!

IP-the-Patent-Guard Intellectual Property

20-year protection
Commercial exclusivity

POWER (The Exclusive-Contract):
Ensures only one Coach can sell the Chemical-Ally, charging high Tolls for Match-Costs.

LORE: The economic engine of drug discovery. Balances Innovation-Reward against Access-Cost.

SCIENCE PUNK!

SUNK-ROCK!

Peer-the-Reviewer-No-2

The Critic

Anonymous
Gatekeeper of the Journal

POWER (The Critical-Block):
Finds every tiny Glitch in the Match-Tape and demands a Remotch (Revision) before the world sees it.

LORE: Quality-Control. The most feared Wrestler in the Academic-Arena.

STAY FLUFFY!

WHAT THE FLUFF?!

KAYFABIZZARO!

Impact-the-Factor-Score Journal Prestige

Citation counts
Hierarchy of trust

POWER (The Prestige-Rank):
Decides how much the world Trusts the results. High-rank journals are the Main-Event-Arenas.

LORE: The Currency-of-Prestige. Leads to hype over steady truth.

HUMBUG!
REJECTED
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KAYFABIZZARO!

MEDKAYFABE!

Doubt-the-Shield-of-Science Skeptical Inquiry

STAT: Nullius in Verba
Tryut the method,
not the person



POWER (The Dogma-Defense): Refuses to believe the Consensus-Wrestler until the Raw-Data-Scrolls are shown.
LORE: Science is the organized skepticism in the reliability of expert opinion.

Surrogate-the-Proxy-Marker Clinical Endpoints

STAT: HbA1c (Diabetes)
Cholesterol (Heart)



POWER (The Number-Game): Focuses on changing a Stat instead of the Real-Outcome (Death). You win the Stat-Match but lose the Life-Match.
LORE: A Better-Number doesn't always mean a Better-Patient.

Correlation-the-Friend NOT Causation

STAT: Co-occurrence
Spurious relationships



POWER (The False-Connection): Shows two things moving together, but warns that one did NOT cause the other.
LORE: The most common Thinking-Trap. Tag-team partners aren't necessarily the boss of each other.

Sensitivity-the-Analysis Robustness

STAT: Changing assumptions
to see if results hold



POWER (The Stress-Tester): Changes the Match-Rules slightly to see if the Verdict still stands. If it breaks, the research was Fragile.
LORE: A strong verdict should survive many different Arena-Conditions.

KayfaBIZARRO!

UN-SCIENCE!

BRAIN-BASH!

BRAIN-BASH!

UN-SCIENCE!

BRAIN-TRAP!

UN-SCIENCE!

DATA-MOSH!

GW

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Citizen-the-Scientist Crowdsourcing

Biohacking
Open-source data



POWER (The People's-Arena): Bypasses the Academic-Monopoly, allowing thousands of Non-Wrestlers to contribute data.
LORE: The democratization of discovery. Patient-Led-Research.

SCIENCE
PUNK!

Real-World-the-Evidence Post-Marketing Surveillance

Phase IV
Observational



POWER (The Wild-Arena-Test): Looks at how the Chemical-Ally performs in the messy real world with Multi-Morb-Max wrestlers.
LORE: Identifies Side-Effect-Saboteurs that only appear once millions are treated.

KAYFABI22ARO!

MEDKAYFABE!

What the
FLUFF?!

Transparen-the-Open-Data Raw Data Sharing

Open data
Reproducible code

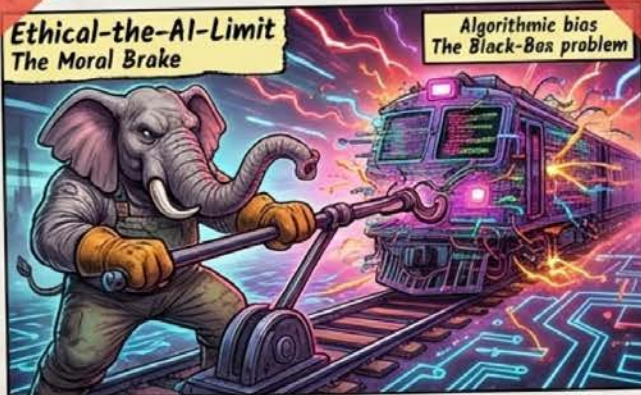


POWER (The Truth-Revealer): Demands the Raw-Match-Tapes be available for anyone to Fact-Check.
LORE: The cure for P-Hacker-the-Cheat. If you show the work, the Cheat can't hide.

DATA-MOSH!

Ethical-the-AI-Limit The Moral Brake

Algorithmic bias
The Black-Box problem



POWER (The Moral-Brake): Ensures the Human-Wrestler still has the Final-Say and isn't just a Data-Point for the AI.
LORE: The challenge of Autonomous-Science. We must ensure the Pattern-Finder is ethical.

KAYFABIZZARO!

MEDKAYFABE!

BRAIN-BASH!

BRAIN-BASH!

DATA-MOSH!

WHAT THE FLUFF?!

The Science Attending Doc FrizzleBob

STAT
Master of Epistemology
Skeptic-in-Chief



POWER (The Rigor-Enforcer)
LORE: Trust the method, not the person.
Doubt the consensus until the data is raw.
Science is a brawl for the truth, not a
parade for the powerful! Stay fluffy!

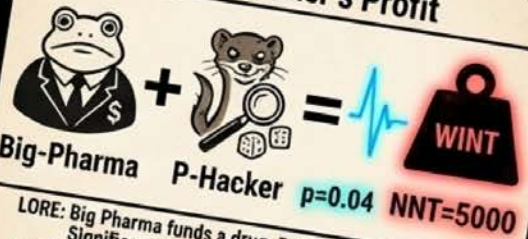


Boss Battle 2 The Blocked Revelation



LORE: A massive Signal challenges the consensus.
The Journal-Fortress refuses publication.
QUEST: Deploy Open-the-Access-Knight and
Pre-Print-Speedrunner to break the siege and
get the data to the arena!

Boss Battle 1 The P-Hacker's Profit



LORE: Big Pharma funds a drug. Results look
Significant, but the NNT is 5000.
QUEST: Identify Scientism-the-False-God and deploy
Popper-the-Falsifier to demand a strict
Replication Match!

Boss Battle 3 The AI-Zebra Hunt



LORE: AI finds a hidden Genetic-Glitch causing
a specific clan to lose Cardio-Matches.
QUEST: Combine Precision-the-Genetic-Sniper
to tailor the drug, while enouring Belmont-
the-Protector ethics are maintained!

Stay
fluffy!